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| **PI Name:** | Perepezko, John Ph.D. | | |
| **Project Title:** | The Dynamics of Heterogeneous Solidification Microstructure Genesis in Undercooled Composite Systems | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | MATERIALS SCIENCE | | |
| **Program/Discipline--Element/Subdiscipline:** | MATERIALS SCIENCE--Basics of solidification | | |
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| **Human Research Program Risks:** | None | | |
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| **Zip Code:** | 53706-1595 | **Congressional District:** | 2 |
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| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 0 |
| **No. of Bachelor's Candidates:** | 2 | **Monitoring Center:** | NASA MSFC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
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| | The ground based program represents a balanced experimental and analysis effort directed toward the investigation of drop tube, glass encased and levitation containerless processing methods. The investigation focus is on the understanding and analysis of microstructural evolution during solidification of undercooled melts and composite systems. The liquid undercooling level in a laboratory scale (3 m) drop tube and levitation melting system can be altered through the variation of processing parameters such as alloy composition, melt superheat, sample size and gas environment. The solidification behavior is evaluated through metallography, SEM, TEM, DTA, and x-ray diffraction in conjunction with a heat flow model of the processing conditions to judge the sample thermal history. A central component of the current program addresses the use of containerless processing together with an undercooled melt with incorporated particles to develop a critical evaluation of solidification front-particle interactions. Another component of the current investigation is the analysis of the control of microstructural evolution in Ni-V, Nb-Pd and Co-Al alloys through the high | | |

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| **Task Description:** | undercooling levels provided by containerless processing to provide fundamental new information on the control of nucleation. The microgravity environment provides new opportunities for the identification and characterization of key parameters that control of solidification processes, and the containerless environment afforded by microgravity conditions avoids crucible-induced sample contamination and allows for large undercooling. The main research objective is the evaluation and analysis of the undercooling and resultant solidification microstructures in containerless processing, including drop tube processing and levitation melt processing of selected alloys and composites. Containerless processing in ground-based drop tubes simulates microgravity conditions via solidification of liquid samples under free fall conditions. The containerless environment is also attained in levitation melt processing and removes a major source of impurities and heterogeneous nucleation sites, allowing for a large melt undercooling. This enhanced liquid undercooling exposes alternate solidification pathways, allowing for the formation of novel microstructures. Controlling the undercooling level provides some control of the operative solidification pathway and the resultant microstructure. The novel structures that may be produced in a ground-based containerless processing facility preview the wide range of possible materials processing experiments that may be conducted in a space-based laboratory. The results of the ground-based study will be used to identify critical experimental variables in microgravity processing and the analysis can be used to design and define the science and hardware requirements for extended duration space experiments. In the solidification of composites the interaction between the solidification front and particles is central to effective processing. Undercooled liquid solidification simulates convection-free solidification conditions for such composites in a 1g environment. Numerous models for the interaction behavior have been proposed for planar solidification fronts, but critical tests are lacking in many cases due to confounding gravitational effects. The interaction of a fast-moving, dendritic solidification front with dispersed particles during solidification processing of metal-matrix composites has revealed a new mode of the incorporation behavior that is not included in existing models for planar interfaces and has not been experimentally accessed previously. In order to complete the analysis of the new microstructural class developed during duplex partitionless solidification it is important to address the key issue of independent or coordinated nucleation of the product phases. Experiments on undercooled samples have provided the microstructural characteristics that are essential to answer this issue. With the critical information on the nucleation mechanism a complete kinetics model analysis can be formulated for the duplex partitionless reaction. The outcome of this program will provide a unique background for a space experiment, because never before has it been possible to quantify the solidification kinetics and the associated microstructure evolution in a manner that can be achieved with the proposed experiments. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | From the previous studies, a set of experiments was focused on the examination of the transitions between particle incorporation modes (i.e. pushing, entrapment, and engulfment). Highly undercooled samples provide an excellent means to study the interaction of high-velocity dendrites and particles and test theoretical models over an extended range of dendrite tip velocities. The studies have shown that the transition from entrapment to engulfment was not observed in the copper-aluminum oxide, copper-zirconium oxide, and nickel-zirconium oxide composite system at any experimentally accessible undercooling (up to 120 K). Samples with copper-nickel alloy matrix materials and zirconium oxide particles at similar undercooling levels (up to 160 K) also do not exhibit engulfment. Furthermore, the experiments have shown that tantalum oxide is of special relevance for the further experimental study of the incorporation behavior since it offers one of the highest densities of the stable oxides and these oxide particles can be engulfed in the matrix at a particular critical interface velocity. Thus, the density difference between the particles and matrix has been identified as a key parameter in prediction the particle incorporation behavior. Nickel-tantalum oxide and copper-tantalum oxide composite systems minimize density driven buoyancy and they can be used to simulate buoyancy-neutral experimental conditions in a terrestrial setting. Current studies reveal that these two composite systems could exhibit large undercooling levels in the presence of crystalline particles. However, the occurrence of particle incorporation behavior is not simply related only to settling. For example, the interface velocity is about 5 orders of magnitude larger than the settling velocity for the composites systems in the current study. Moreover, calculations based on the current experimental results have demonstrated the importance of the lateral growth of the dendrite trunk and the growth of dendrite side branches in altering the particle incorporation behavior. The effects of other experimental factors need to be considered in order to evaluate more complete analytical models.<br>Experiments are currently in progress to determine the effect of particle size and the distribution of particles on the microstructure of nickel-tantalum oxide and copper-tantalum oxide composite systems. The current studies have established that large undercooling levels can be achieved in the model composite systems and that these undercooling levels depend on the particle size. The samples studied in this work undercool reproducibly by more than 100 K. The particles with an average diameter larger than 10 micron are confined around the sample surface following solidification at all undercooling levels. Experimental analyses have shown that the particles are distributed homogeneously inside the liquid sample. Thus, the surface confinement of particles is a result of particle movement produced by the moving solidification front. Furthermore, the particle size also alters the mechanism of grain refinement in an undercooled liquid. Detailed studies of the effects of interface velocity and alloy composition including the composition that yields neutral buoyancy for particles and the effect of other experimental factors will be conducted to yield a clear understanding of the particle/solidification-front interaction. Studies on the effect of particle morphology and convective fluid flow on incorporation behavior in the matrix are also underway.<br><br>Currently, a model to simulate DTA signals has been developed for the melting and dendritic solidification of an undercooled pure material. The model calculations are in qualitative agreement with experimentally measured melting and freezing DTA signals. With a more sensitive DTA apparatus, it may be possible to use this DTA model to determine solidification velocity by fitting the experimental data to a simulated curve.<br><br>The current studies will provide a database for future space experiments where the incorporation behavior can be examined in the absence of the confounding influence of buoyancy-induced convection. The combination of ground-based studies and experiments under microgravity conditions will provide an understanding of the key parameters of the particle incorporation process that can also lead to improved composite processing strategies for both uniform gradient particle distributions. |
| **Bibliography Type:** | Description: (Last Updated: 04/29/2010) |

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| **Abstracts for Journals and Proceedings** | Perepezko JH. "Phase Selection Transitions During Udercooled Melt Solidification." Invited talk, 33rd Annual Meeting & Exhibition of The Minerals, Metals & Materials Society (TMS), Charlotte, NC, March 14-18, 2004.<br>33rd Annual Meeting & Exhibition of The Minerals, Metals & Materials Society (TMS), Charlotte, NC, March 14-18, 2004. , Mar-2004 |
| **Books/Book Chapters** | Perepezko JH. "Phase Selection Transitions During Undercooled Melt Solidification." in "Solidification Processes and Microstructures-A Symposium in Honor of Wilfried Kurz." Ed. M. Rappaz, C. Beckermann, and R. Trivedi. Wiley, 2010., Apr-2010 |